I Finite Automaton / Finite State Machine

Main Idea: Use FA to select which vector field to use

- known as hybrid control
- Can be "fancier" by including the dynamics into the definition of the automation

A) Definition

A Finite Automaton (FA) is a 5-tuple $(Q, \Sigma, q_0, A, \delta)$ where

Q : A finite set of states

Σ : A finite input alphabet

$q_0 \in Q$: Initial State

$A \subseteq Q$: Set of accepting States

$\delta : Q \times \Sigma \to Q$: Transition function

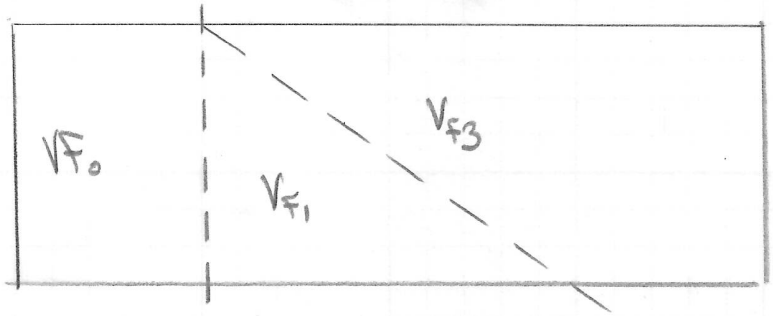
For any element q of Q any any symbol $\sigma \in \Sigma$, we interpret $\delta(q, \sigma)$ as the state to which FA moves, if it is in state q and receives input σ.

II Create a Robust Switching Distance

A) Regions

A typical (easy) way to implement a switching distance is when we are with a radius of a desired position. However, if other factors (such as an orbit to avoid an object) prevents us from entering the radius we will never switch.

↳ Solution, define regions for certain fields



Define dividing lines perpendicular to line created by vector field.

Point on perpendicular line: x_{l_i}

Orientation perpendicular line: $\psi_{l_i} \pm \pi/2$

Form function:

$$f_i(x,y) = a_i x + b_i y + c_i$$

National® Brand 42-381 50 SHEETS EYE-EASE® - 5 SQUARES 42-382 100 SHEETS EYE-EASE® - 5 SQUARES 42-389 200 SHEETS EYE-EASE® - 5 SQUARES